

Human-Computer Interaction

Generative AI, Creativity & Tools for Thought

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Today's Agenda

- Topic overview: *Generative AI, Creativity & Tools for Thought*
- The frame: *does the tool augment the thinker, or substitute for the thinking?*
- Provocation: *Does GenAI grow or diminish human thinking?*
- Discussion

This term's frame

Enduring fundamental → How AI is changing it → Depth (a claim to argue with)

- **Fundamental:** computing as an *intellectual amplifier*, extending thought (Bush, Engelbart)
- **How AI changes it:** generation makes the **artifact** cheap; the scarce skills become *steering* and *verification*
- **Depth: Deep:** it questions what the tool is for, and starts the *deskilling* thread

This week's readings

- **Required:** Lee et al., *The Impact of Generative AI on Critical Thinking* (CHI 2025)
- **Recommended:** Bush, *As We May Think* (1945), re-read from Week 3
- **Optional:** Dhillon et al., *Varied Scaffolding Levels in Co-writing with Language Models* (CHI 2024)

Topic overview: ***Generative AI, Creativity & Tools for Thought***

The lineage: tools for thought

Two ideas, both about making the human more capable:

Externalize (*Bush, 1945*)

Store and link thought outside the head: "an enlarged intimate supplement to his memory"

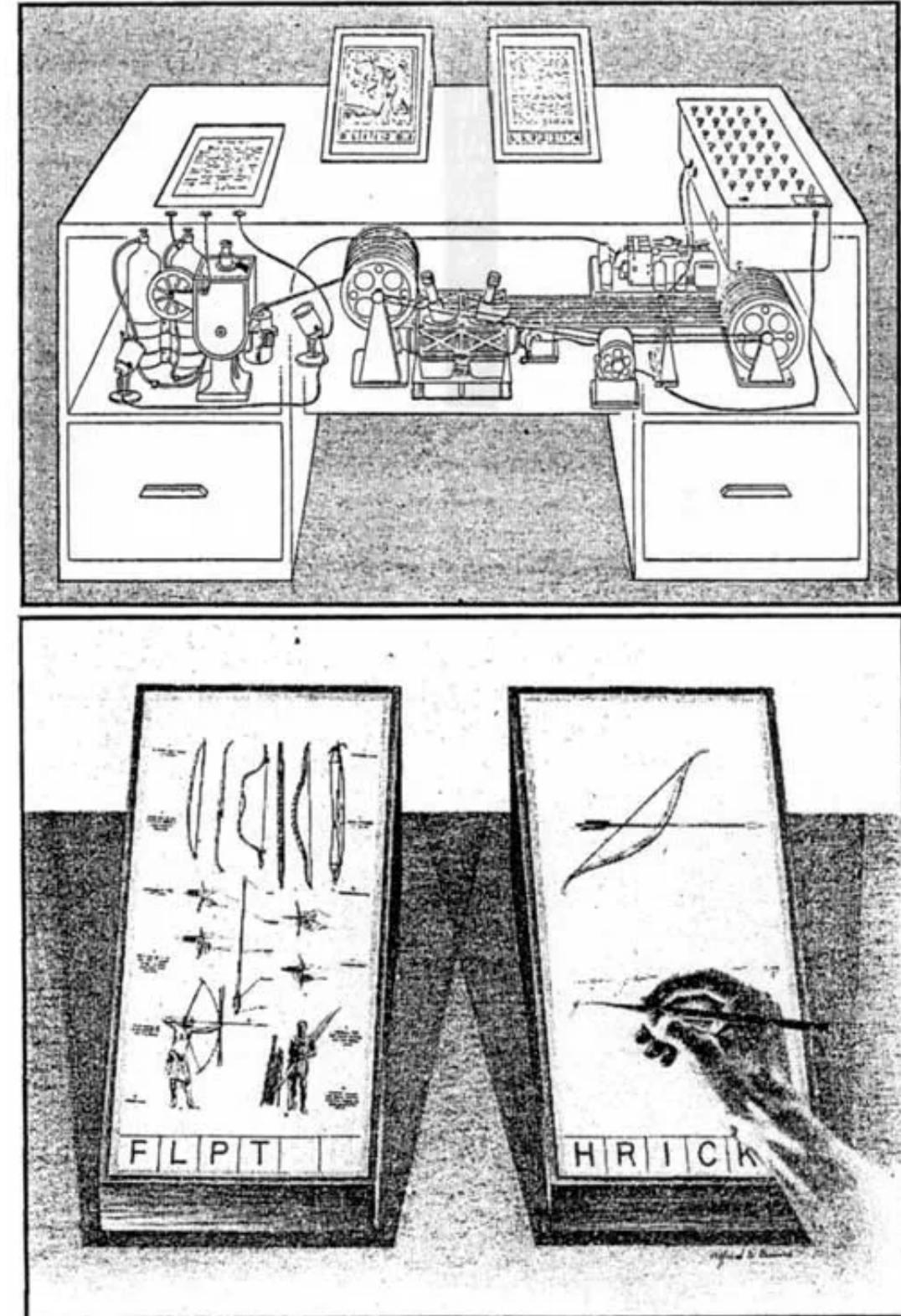
Augment (*Engelbart, 1962*)

"increasing the capability of a man to approach a complex problem situation"¹

¹ Engelbart, D. (1962). *Augmenting Human Intellect: A Conceptual Framework*. SRI Summary Report AFOSR-3223, §1a.

1945: Bush's memex²

- A desk that stores "all his books, records, and communications"
- Organized the way the mind works: it "operates by association"
- The line he drew: "For mature thought there is **no mechanical substitute.**"



² Bush, V. (1945). [As We May Think](#). *The Atlantic*.

Distributed cognition³

Thinking isn't confined to "the skin or skull of an individual." It is spread:

- **across people**: a crew, a team
 - **between inside and outside**: notes, instruments, tools
 - **through time**: earlier work reshapes later work
- 👉 External aids don't just add memory; they **change the task**.

³ Hollan, Hutchins & Kirsh (2000), *ACM TOCHI*, 7(2), 174-196.

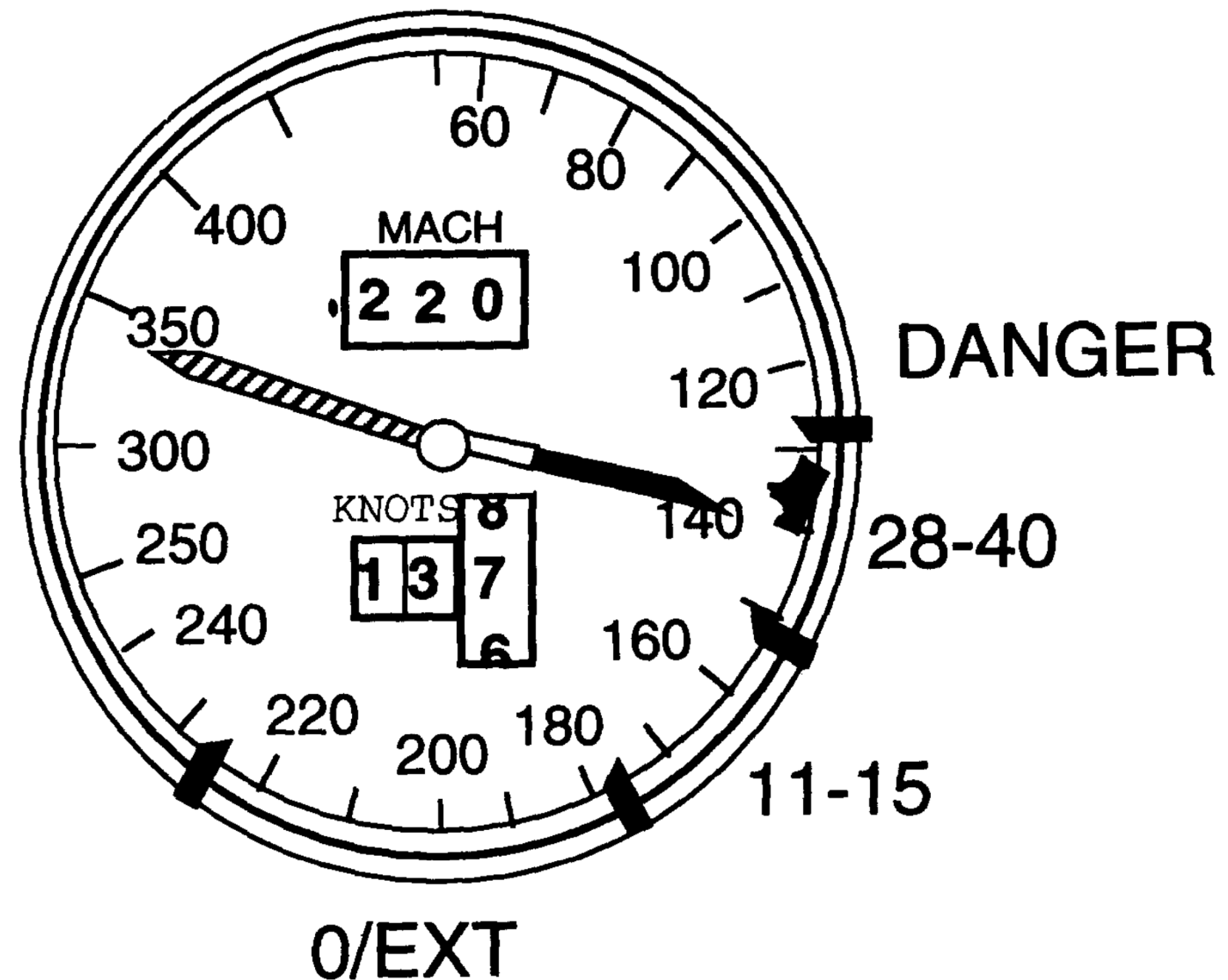
How a cockpit remembers its speeds⁴

Pilots set **speed bugs** on the airspeed dial before landing.

- Worked out early, read later at a glance
- Memory becomes perception: is the needle at the bug?

"Speed bugs do not help pilots remember speeds; rather, they are part of the process by which the cockpit system remembers speeds."

⁴Hutchins (1995), *Cognitive Science*, 19(3). Fig. 4.



What AI changes: the cost of the artifact

Engelbart's 1962 "writing machine" made drafts cheap, so the writer could think more.

Generative AI makes the artifact cheap by **writing it for you**.

- The bottleneck moves from *making the thing* to *knowing what to make, and whether it's good*
- New core skills: **steering** (what to ask for, keep, and change) and **verification**

👉 Cheap output is not the same as understanding.

It grades its own options, too

A "**Recommended**" tag hands over the other half: *whether it's good*.

- **Defaults stick:** the preselected option tends to win⁵
- **Suggestions narrow:** shorter, more predictable writing⁶
- **Suggestions persuade:** users' own views shifted⁷
- **Advice gets followed,** even when it's wrong⁸

⁵ Johnson & Goldstein (2003), *Science*.

⁶ Arnold, Chauncey & Gajos (2020), *IUI '20*.

⁷ Jakesch et al. (2023), *CHI '23*.

⁸ Parasuraman & Manzey (2010), *Human Factors*.

Designing against it

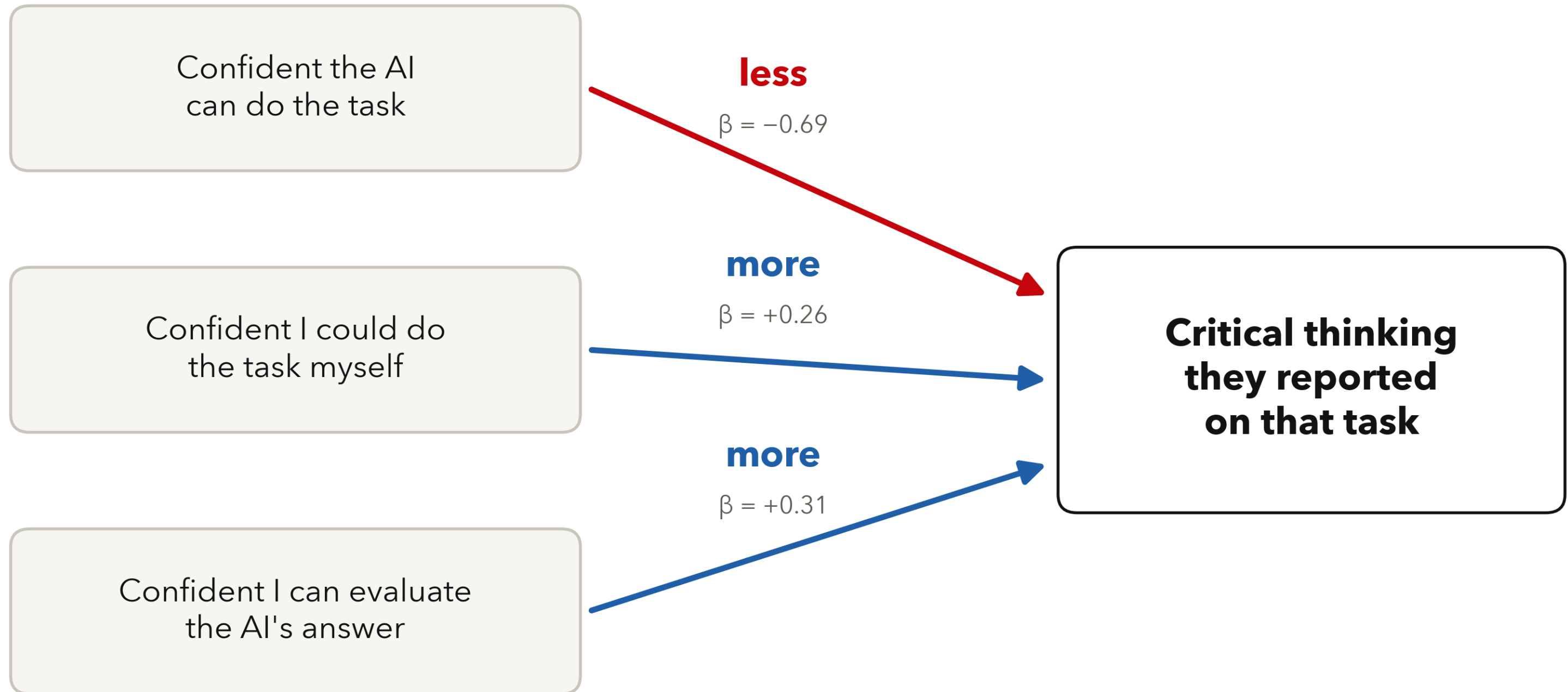
- **Decide first, then see the AI:** cognitive forcing reduced overreliance, though people liked it least⁹
 - **AI that challenges, not obeys:** a "Socratic gadfly" that questions your choices instead of making them¹⁰
- 👉 Design test: does the tool leave room for *your* judgment before it offers its own?

⁹ Buçinca, Malaya & Gajos (2021), CSCW (Week 4).

¹⁰ Sarkar, A. (2024). AI Should Challenge, Not Obey. *CACM*, 67(10), 18-21.

Confidence and critical thinking

Lee et al. (2025): 319 knowledge workers, 936 real uses of GenAI at work



Each task rated by the person who did it. All self-reported; correlations, not causes.

Lee et al. (2025), CHI '25, §4.2.1: mixed-effects regression on perceived enactment of critical thinking.

The central tension

Bush drew the line: "For mature thought there is no mechanical substitute."

Augment

The tool takes the repetitive part,
and you end up **more capable**
(Engelbart)

Substitute

The tool takes the thinking itself;
without practice, the skill can
erode (Bainbridge, via Lee et al.)

How much should the tool write?¹¹

131 people wrote with no AI, with next-sentence, or with next-paragraph suggestions.

- **Paragraphs:** better quality and productivity, most for less regular writers
- **Sentences:** no significant gain
- **Both:** lower ownership and satisfaction; same workload

👉 Not "less is better": the help that raised quality lowered ownership.

¹¹ Dhillon, P.S., Molaei, S., et al. (2024). Shaping Human-AI Collaboration. *CHI '24*. doi:10.1145/3613904.3642134

Can you verify what the tool shows you?

Lee et al.: with GenAI, critical thinking shifts toward **verification**.

- Tools label their answers: high, medium, or low confidence
- But a label is itself a claim, and it can be wrong
- One answer can't tell you: a "medium" answer is *supposed* to be wrong sometimes

👉 The only check is the tally: of all the "medium" answers, how many were right?

You can only check a label in aggregate

40 answers from an AI tool, tallied by confidence label:

Label	Claims	Actually right
High	85% or more	18 of 20 (90%)
Medium	70-84%	4 of 10 (40%)
Low	under 70%	6 of 10 (60%)

👉 "Medium" was wrong most of the time; "Low" did better.

Guardrails coda

Three questions to carry into any creativity tool:

- **Attribution / authorship**: who is the author when the artifact is generated?
- **Over-reliance**: does fluent output inflate confidence past competence?
- **Whose labor**: whose work did the corpus *launder* to make this cheap?

**But what about the
thinker?**

The dialectic, today

Generative AI is the newest move in the tools-for-thought lineage.

- **Bush, Engelbart:** augment the mind, don't spare it
 - **Lee et al.:** trust in the tool, less reported critical thinking
 - **Dhillon et al.:** more help, better text, less ownership
- 👉 Design test: are you a **better thinker** when it's off?

Your provocation (*post one slide*)

Pick one:

1. **Provocation**: does generative AI *grow* or *diminish* human thinking? Ground it in the readings.
2. **Critical artifact**: a real GenAI tool that makes the *steering* / *authorship* / *deskilling* tension visible
3. **Design / policy**: one guideline that keeps the human the *thinker*, not the spectator

Discussion

Today's roles

Reporter

Skeptic

Connector

Seat 4

Seat 5

Seat 6

Same in every pod. **Seats 1, 2 and 3** have no role today; you are still in the argument.

Discussion Format

In your pod, on your pod monitor: ~30 minutes.

- **Everyone shows their slide:** about 2 minutes each, cast from your own laptop
- **Skeptic** puts a counterargument on the table · **Connector** ties it to an earlier week
- **Reporter** writes the pod's three lines (agreed, split, unresolved) and delivers them to the room

What's Next?

- **Friday: Experimental Design + Measurement (M5)**
 - **Read:** LFH Ch. 2-3
 - Post your team's 3-line project status
- **Due Fri Oct 9, 9:30 am:** Qualitative Field-Method Plan (*individual*)